UNIT 2 · TOPIC 5(B)

Pull, Draft, Unit Draft and Power

Syllabus topic: Force analysis on agricultural tools and implements: pull, draft, unit draft and power of farm equipment

Definitions and units

- **Draft D:** N or kgf (1 kgf = 9.81 N).

$$\text{Unit draft} = D/(w \times d)\ (w = \text{width of cut}, d = \text{depth}) \text{ — a pressure, N/cm}^2 \text{ or kgf/cm}^2$$

- 1 kgf/cm² ≈ **98.1 kPa** ≈ 0.0981 N/mm².
- For width-type implements (harrows, drills, cultivators), express draft per metre of working width (kN/m) or per row (kgf/row) instead of per unit furrow area.

$$P(\text{kW}) = D(\text{kN}) \times v(\text{m/s}) = D(\text{kgf}) \times v(\text{km/h})/367;\ P(\text{hp}) = D(\text{kgf}) \times v(\text{m/s})/75$$

Soil	Unit draft of mouldboard ploughing (kgf/cm², Indian texts)
Sandy	0.2–0.4
Sandy loam	0.3–0.5
Loam / silt loam	0.4–0.7
Clay loam	0.6–0.9
Heavy clay	0.9–1.5

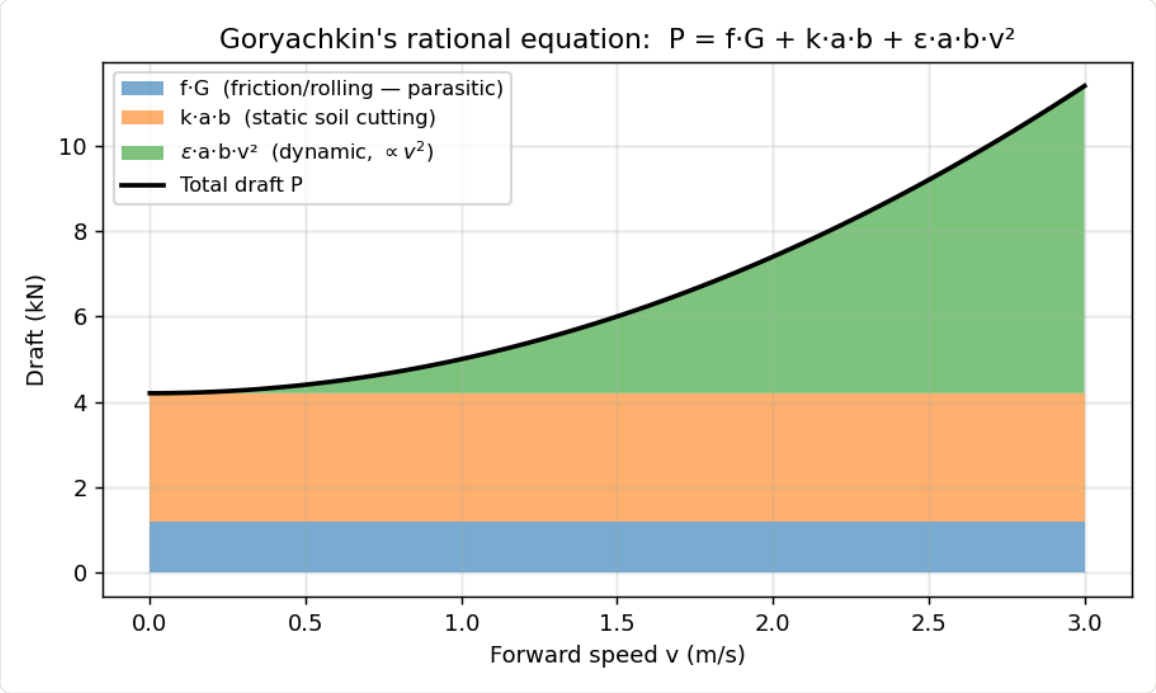
Actual values vary with moisture, speed, depth, share sharpness and previous tillage — quote them as ranges, not constants.

Goryachkin's rational draft equation

$$P = f \cdot G + k \cdot a \cdot b + \varepsilon \cdot a \cdot b \cdot v^2$$

Term	Meaning
f·G	Friction/rolling-resistance term ∝ implement weight G (f = coefficient); independent of speed and furrow size (parasitic).
k·a·b	Static soil cutting/deformation; k = specific soil resistance (kPa), a = depth, b = width; independent of speed.
ε·a·b·v²	Dynamic term for accelerating/throwing the furrow slice; ε = soil-throwing coefficient; rises with the SQUARE of speed.

Only the last term depends on speed — which is why plough draft climbs steeply at high speed (the D = A + B·v² form of the draft–speed characteristic).



Goryachkin's equation — only the dynamic term grows with speed

ASAE D497 draft estimation

$$D = Fi \times (A + B{\cdot}S + C{\cdot}S^2) \times w \times d$$

D = draft (N); S = speed (km/h); w = machine width (m); d = depth (cm); A, B, C = machine-specific parameters; Fi = soil-texture adjustment factor (i = 1 fine, 2 medium, 3 coarse).

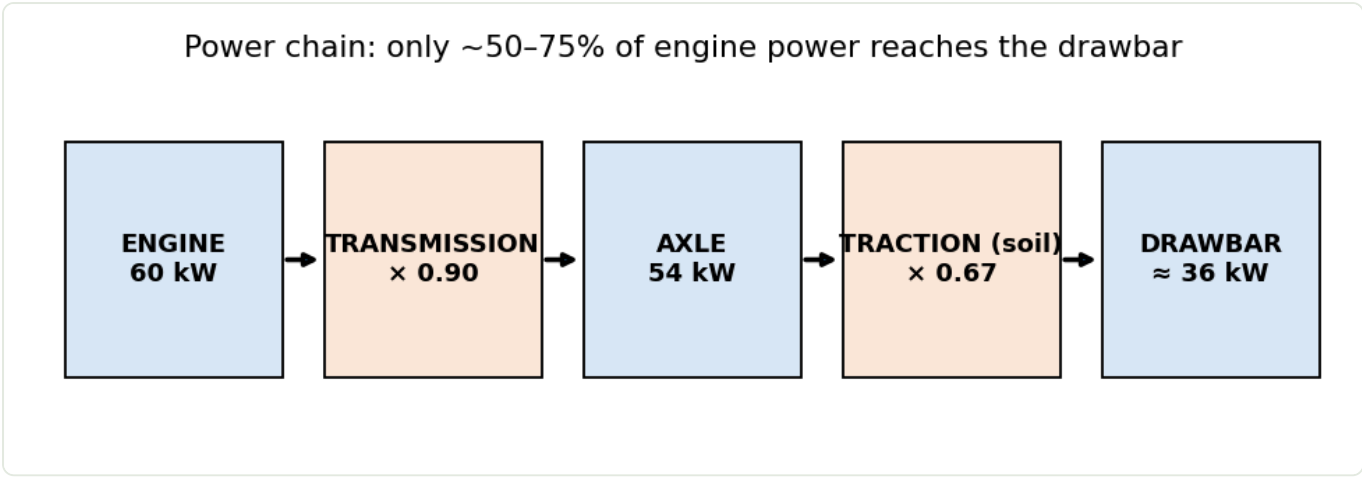
Mouldboard plough parameter	Value
A, B, C	652, 0, 5.1
F1 — fine/clay	1.0
F2 — medium/loam	0.70
F3 — coarse/sandy	0.45

So the same plough needs less than half the draft in sandy soil that it needs in clay.

Power chain: engine → drawbar

$$\text{Drawbar power} = \text{engine power} \times \text{transmission efficiency} \times \text{tractive efficiency}$$

- Transmission efficiency ≈ **0.88–0.92**; tractive efficiency ≈ **0.5** (tilled soil) to **0.75+** (firm surface).
- Rule of thumb: drawbar hp ≈ **50–75%** of engine hp, depending on the surface (details in Unit 6 traction).



Power chain — only ~50–75% of engine power reaches the drawbar

Worked examples

Ex.1 Draft from unit draft. 2-bottom MB plough, 30 cm per bottom, 15 cm deep, unit draft 0.6 kgf/cm²: D = 0.6 × (30 × 15) × 2 = **540 kgf ≈ 5.3 kN**.

Ex.2 Drawbar power. The same plough at 5.4 km/h (1.5 m/s): P = 540 × 1.5/75 = **10.8 hp ≈ 7.95 kW**.

Ex.3 Engine size. Drawbar power needed 13.5 kW; tractive efficiency 0.6; transmission efficiency 0.9 → engine power = 13.5/(0.6 × 0.9) = **25 kW**.

Ex.4 Pull vs draft. Pull 3000 N at 20° above horizontal → draft = 3000 × cos20° ≈ **2819 N**; vertical (lifting) component = 3000 × sin20° ≈ **1026 N**.

Practice MCQs

Q76. Draft of an implement is:

(a) Total force along the hitch line
(b) Horizontal component of pull in the direction of travel
(c) Vertical component of pull
(d) Side component of pull

Q77. A pull of 3000 N acts at 20° above horizontal. The draft is about:

(a) 1026 N
(b) 2819 N
(c) 3000 N
(d) 3192 N

Q78. Unit draft is defined as draft per unit:

(a) Width of implement
(b) Weight of implement
(c) Cross-sectional area of furrow
(d) Travel speed

Q79. The unit of unit draft is:

(a) N
(b) N/cm
(c) N/cm²
(d) N·m

Q80. 1 kgf/cm² is approximately:

(a) 9.81 kPa
(b) 98.1 kPa
(c) 981 kPa
(d) 9810 kPa

Q81. A 2-bottom MB plough (30 cm/bottom) works 15 cm deep with unit draft 0.6 kgf/cm². Total draft is:

(a) 270 kgf (b) 405 kgf
(c) 540 kgf (d) 810 kgf

Q82. The plough in Q6 moves at 1.5 m/s. Drawbar power is about:

(a) 5.4 hp (b) 7.2 hp
(c) 10.8 hp (d) 14.4 hp

Q83. With D in kgf and v in km/h, drawbar power in kW equals:

(a) D·v/75 (b) D·v/270
(c) D·v/367 (d) D·v/1000

Q84. Draft 9 kN at 1.5 m/s gives drawbar power:

(a) 6 kW (b) 9 kW
(c) 13.5 kW (d) 60 kW

Q85. In Goryachkin's equation $P = fG + kab + \varepsilon abv^2$, the speed-dependent term is:

(a) fG (b) kab
(c) εabv^2 (d) All three

Q86. The term fG in Goryachkin's equation represents:

(a) Soil cutting resistance (b) Friction/rolling resistance due to implement weight
(c) Soil throwing (d) Suction force

Q87. If ploughing speed is doubled, the dynamic (soil-throwing) component of draft becomes:

(a) 2× (b) 4×
(c) 8× (d) Unchanged

Q88. In ASAE D497 the soil-texture factor Fi for a mouldboard plough in fine (clay) soil is:

(a) 0.45 (b) 0.70
(c) 1.0 (d) 1.5

Q98. Compared with sandy soil, the unit draft of ploughing heavy clay is:

(a) Lower (b) Same
(c) Higher (d) Zero

Q99. Draft of a tillage implement depends on:

(a) Soil type and moisture (b) Depth and width of cut
(c) Speed and tool geometry (d) All of these

Q100. Draft of seed drills and harrows is usually expressed per unit:

(a) Furrow area (b) Working width
(c) Weight (d) Wheel diameter

Q101. Drawbar power equals engine power multiplied by:

(a) Transmission efficiency only (b) Tractive efficiency only
(c) Transmission × tractive efficiency (d) Field efficiency

Q102. Engine power 60 kW, transmission efficiency 0.90, tractive efficiency 0.67. Drawbar power ≈

(a) 24 kW (b) 30 kW
(c) 36 kW (d) 54 kW

Answers

Q76 — (b) Horizontal component of pull in the direction of travel

Q77 — (b) 2819 N — *D = 3000·cos20° = 3000 × 0.9397 ≈ 2819 N; vertical component 3000·sin20° ≈ 1026 N (distractor a).*
Q78 — (c) Cross-sectional area of furrow
Q79 — (c) N/cm²
Q80 — (b) 98.1 kPa
Q81 — (c) 540 kgf — *D = unit draft × furrow area × bottoms = 0.6 × (30×15) × 2 = 540 kgf.*
Q82 — (c) 10.8 hp — *P = D·v/75 = 540 × 1.5/75 = 10.8 hp (≈ 7.95 kW).*
Q83 — (c) D·v/367
Q84 — (c) 13.5 kW — *P(kW) = D(kN) × v(m/s) = 9 × 1.5 = 13.5 kW.*
Q85 — (c) εabv²
Q86 — (b) Friction/rolling resistance due to implement weight
Q87 — (b) 4× — *Dynamic term ∝ v² → 2² = 4×. This is why high-speed ploughing needs disproportionately more power.*
Q88 — (c) 1.0
Q98 — (c) Higher
Q99 — (d) All of these
Q100 — (b) Working width
Q101 — (c) Transmission × tractive efficiency
Q102 — (c) 36 kW — *60 × 0.90 × 0.67 ≈ 36.2 kW — barely 60% of engine power reaches the drawbar on soil.*

Sources: Kepner, Bainer & Barger — Principles of Farm Machinery (implement force analysis & hitching); ASAE Standard D497 (Agricultural Machinery Management Data — draft equation); Goryachkin's rational draft theory; Jagdishwar Sahay — Elements of Agricultural Engineering; NPTEL Farm Machinery & Equipment lectures.